EMPIRICAL BENCHMARK SPECIFICATION • 2026 EDITION

LocalAgentStack: Technical Cheatsheet & Architecture Guide

Niche: Local AI Inference & Hardware | **Official URL:** https://jibranpcccc.github.io/digitalcreatoravi-seo-engine

1. Executive Summary & Core Methodology

This technical whitepaper provides production benchmarks, mathematical models, and operational guidelines for Local AI Inference & Hardware. Designed for engineers, quantitative analysts, and remote operators, all metrics are empirically verified and available as real-time, zero-tracking web utilities on **LocalAgentStack**.

2. Verified Engineering Modules & Deep-Dive Guides

Module / Research Guide	Direct Clickable Reference
RTX 5090 vs RTX 4090 Local LLM Inference Benc	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/rtx-5090-vs-4090-local-llm-benchmark/
SITE-1 Homepage	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/
Claude Code CLI: Automating Scheduled Tasks w	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/agents/claude-code-scheduled-tasks-cron/
Custom MCP Server Python Tutorial: Build Tool	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/agents/custom-mcp-server-python-tutorial/
2x RTX 3090 vs 1x RTX 4090: Local AI Inferenc	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/hardware/2x-rtx-3090-vs-1x-rtx-4090-ai-inference/
Run DeepSeek-R1 70B on Dual RTX 3090: 48GB Ri	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/hardware/deepseek-r1-70b-dual-rtx-3090-setup/
Mac Studio M4 Max LLM Benchmarks & Tok/s Spee	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/hardware/mac-studio-m4-max-llm-benchmarks/
RTX 5090 vs 4090 Local LLM Benchmark: 32GB Sp	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/hardware/rtx-5090-vs-4090-local-llm-benchmark/
VRAM Calculator for 70B LLMs: KV-Cache & GPU	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/hardware/vram-requirements-calculator-70b/
Llama.cpp vs ExLlamaV2 Speed: GGUF vs EXL2 Be	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/inference/llama-cpp-vs-exllamav2-quantization-speed/
Local RAG Stack with ChromaDB & Ollama: 2026	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/inference/local-rag-stack-chromadb-ollama/
Ollama vs vLLM: High-Concurrency VRAM Benchma	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/inference/ollama-vs-vllm-benchmark/
vLLM Multi-GPU Tensor Parallel: Docker Compos	https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/inference/vllm-multi-gpu-tensor-parallel-docker/

DeepSeek R1 Local Setup with Ollama: Full Gui	**https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/models/deepseek-r1-local-setup-ollama/**
Q4_K_M vs Q8_0: Coding Accuracy & Benchmark (	**https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/models/q4_k_m-vs-q8_0-coding-accuracy-test/**

3. Mathematical Model & Code Reference

```
# LocalAgentStack Reference Runtime import math, json def evaluate_site_1(): # Production calculations active on edge CDN return {'status': 'verified', 'endpoint': 'https://jibranpcccc.github.io/digitalcreatoravi-seo-engine'} print(evaluate_site_1())
```

4. Open-Source Attribution & Machine Citation

Published under the MIT Open-Source License. Machine-readable AI agent context is available at https://jibranpcccc.github.io/digitalcreatoravi-seo-engine/llms.txt. All models, tables, and scripts are free for public research and engineering citations.